

Indian Institute of Technology Gandhinagar

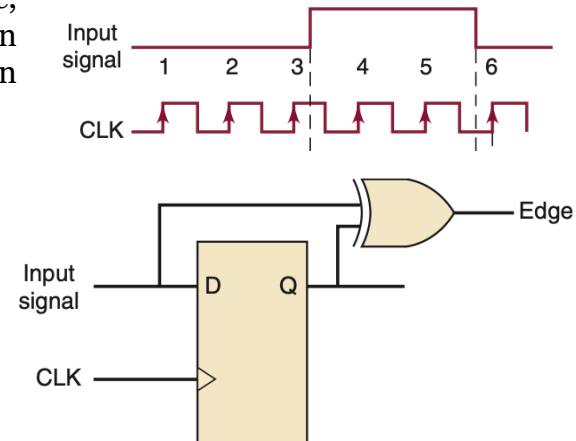
ES116 Principles and Applications of Electrical Engineering

Semester II, 2025-2026

Tutorial-12 (Sequential Circuits)

Q1. You are analyzing the circuit shown in the figure, which feeds an Input Signal into a D FF and then compares that Input Signal with the output (Q) using an XOR gate.

- Draw the output of the XOR gate (Edge) for the given Input signal and CLK. Comment on its functionality.
- Now, assume you replace the Flip-Flop with a D Latch, and you connect the CLK signal to the Enable Pin of a D Latch. Draw the output of the XOR gate and comment on its differences from part A.



Q2. You have a single positive-edge triggered D Flip-Flop. You wire the inverted output ($\bar{Q}$) directly back into the D input. If you connect a clock signal (as shown in Question 1) to the clock pin, draw the output Q of the DFF? Comment on the frequency of CLK and obtained Q signal.

Q3. You are given the conversion table as shown here. To build the D Flip Flop using S-R FF,

- Fill the required S and R inputs based on Q_n and Q_{n+1} . (i.e. Excitable Table for SR FF)
- Use K-maps to find S and R expression in terms of D and Q_n .

Input	Present state	Next state	Flip-flop inputs	
D	Q_n	Q_{n+1}	S	R
0	0	0		
0	1	0		
1	0	1		
1	1	1		

- Draw the resultant logic circuit using SR FF.
- Derive the expressions for S and R if we want to implement a transparent D latch using an SR latch with an enable input E.

Q4. You are designing a logic circuit for a simple cargo elevator that serves only two floors: Floor 0 (Ground) and Floor 1 (Top). State $Q=0$ means the elevator is at Floor 0. State $Q=1$ means the elevator is at Floor 1. There is a single "Move" button, labeled M. If $M=0$, the elevator stays where it is. If $M=1$, the elevator moves to the *other* floor.

- Draw the state transition diagram (circles for states, arrows for transitions based on M).
- Fill out the State Transition Table.
- Derive the Boolean logic equation for the Flip-Flop input D in terms of current state Q and input M. What basic logic gate does this represent?

Q5. Design a simple 2-bit synchronous counter that continuously counts upward: $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$, using D Flip-Flops.

- A. Write the State Transition Table showing Current State (Q_1, Q_0) and Next State (D_1, D_0).
- B. Find the simplified Boolean logic equation for D_0 .
- C. Find the simplified Boolean logic equation for D_1 .

Q6. The campus has installed a new automated Chai dispenser. A cup of chai costs exactly 20 Rupees. The machine only accepts 10-Rupee coins.

- A. To save hardware costs, your institute decides to install an "instant" dispenser. The exact moment the second 10-Rupee coin is inserted, it immediately pours the chai and resets. It never "waits" with 20 Rupees inside. Input X: $X=1$ for exactly one clock cycle when a 10-Rupee coin is inserted ($X=0$ otherwise). Output Y: $Y=1$ triggers the dispenser to pour the chai ($Y=0$ keeps it locked). Use exactly one D Flip-Flop. State 0 ($Q=0$) means 0 Rupees inserted. State 1 ($Q=1$) means 10 Rupee inserted.

1. Draw the State Transition Diagram (using arrows labeled X/Y).
2. Fill out the State Transition Table.
3. Derive the simplified logic equations for the Next State (D) and Output (Y).
4. Sketch the final gate-level circuit.

- B. Students complained that the "instant" dispenser from Part A spilled chai on their hands because it poured too fast! You are tasked with upgrading the machine. The machine must now hold the 20 Rupees and wait for the user to press a button before pouring. Input $X=1$ when a 10-Rupee coin is inserted, $P=1$ when the "Pour" button is pressed. (Assume X and P are never pressed at the exact same time). Output Y: $Y=1$ pours the chai. The machine now requires three states: S_0 (00): 0 Rupees, S_1 (01): 10 Rupee, S_2 (10): 20 Rupees (Ready to pour), and S_3 (11): Invalid / Don't Care.

1. Draw the new State Transition Diagram. Include states S_0, S_1 , and S_2 . Show what happens if X is inserted, and what happens if P is pressed. (*Hint: If P is pressed while in S_0 or S_1 , the machine should just stay in that state*).
2. Create the State Transition Table. There are 2 Flip-Flops (Q_1, Q_0) and 2 External Inputs (X, P).
3. Derive the logic equations for the Next States (D_1, D_0) and Output (Y).

(Optional Practice Question) Q7. Design a synchronous BCD (Binary-Coded Decimal) up-counter using positive-edge triggered D Flip-Flops. The counter must sequence from 0000 to 1001 and roll over to 0000 on the next clock cycle.

- A. Construct the complete table detailing Current State (Q_3, Q_2, Q_1, Q_0) and Next State (D_3, D_2, D_1, D_0). Explicitly define states 1010 through 1111 as "don't care" (X) conditions.

- B. Draw and solve four separate 4-variable K-maps to derive the inputs for D3, D2, D1, and D0. Show all your groupings.
- C. Provide the final, maximally simplified Boolean sum-of-products equations for all four Flip-Flop inputs.
- D. Sketch the final hardware implementation showing the four D Flip-Flops, the clock routing, and the combinational logic gates.